

STII

## Defence Technology

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*About 2-4 questions a paper. DRDO facts and the exercise-to-country match are the two near-certain items; missile ranges and new commissionings supply the rest.*

DRDO and its products have appeared in every RAS paper since 2016 - the founding year, the motto and Lukoskin are documented repeats. Around them sit the missiles with their ranges, the platforms of the three services, the nuclear doctrine, and the joint exercises matched to their partner countries.

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## India's Defence Technology

### DRDO

- Founded 1 January 1958
- Balasaya Mulam Vigyanam
- About 52 laboratories
- Lukoskin, 2-DG, rations
- Defence Laboratory Jodhpur

### Missiles

- IGMDP 1983 - Prithvi, Agni, Trishul, Akash, Nag
- BrahMos, Astra, Nirbhay
- QRSAM, VSHORADS, Pinaka, Pralay
- K-series SLBM
- Agni-V MIRV, HSTDV

### Air

- Tejas LCA, AMCA
- Rafale, Su-30MKI
- LCH Prachand, Dhruv
- TAPAS-BH, Netra AEW&C

### Sea

- INS Vikrant, Vikramaditya
- Arihant, Arighaat SSBN
- Kalvari class, Project 75
- Project 15B and 17A

### Land

- Arjun MBT, Zorawar
- ATAGS, Dhanush, K-9 Vajra
- Pinaka MBRL
- AK-203 at Korwa

### Nuclear

- Pokhran-I 1974
- Pokhran-II 1998
- Doctrine 2003 - No First Use
- Triad completed 2018
- MTCR, Wassenaar, Australia Group

### Structure and industry

- CDS and DMA 2019
- Andaman and Nicobar Command 2001
- Agnipath 2022
- OFB into 7 DPSUs 2021
- Defence corridors UP and Tamil Nadu
- Joint exercises

**DRDO - the organisation RPSC asks about every year**

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DRDO is the research and development wing of the Ministry of Defence. Its job is not to manufacture in bulk - that is for the DPSUs and private industry - but to design systems and hand the technology over. It began small and became the largest research network in the country. These founding facts are documented repeats: RPSC has asked the year, the motto and one product in paper after paper.

- Founded on 1 January 1958 by merging the Technical Development Establishment, the Directorate of Technical Development and Production, and the Defence Science Organisation.
- Started with 10 laboratories; today about 52.
- Motto: 'Balasya Mulam Vigyanam' - 'Strength's origin is in Science', drawn from Kautilya's Arthashastra.
- Headquarters: DRDO Bhawan, New Delhi.
- Works under the Department of Defence Research and Development in the Ministry of Defence.
- Its head is Secretary, Department of Defence R&D and Chairman, DRDO.
- Five DRDO Young Scientist Laboratories were opened in 2020 - artificial intelligence, quantum technologies, cognitive technologies, asymmetric technologies and smart materials.
- Aero India, the biennial air show, is held at Yelahanka, Bengaluru.

#### ASKED IN RAS

Two near-certain repeats. RAS Pre 2024 asked the year DRDO was established - 1958. RAS Pre 2018 asked the motto - Balasya Mulam Vigyanam. If you learn nothing else in this chapter, learn these two.

Now the laboratories. The pattern to hold is city-to-subject, because that is how the match-the-following is framed. Hyderabad means missiles, Bengaluru means aeronautics and radar, Pune means armaments, Visakhapatnam and Kochi mean underwater, Mysuru means food.

### DRDO's laboratories - city to subject



### DRDO laboratories and their specialities

| Laboratory                                             | Location       | Area of work                                       |
|--------------------------------------------------------|----------------|----------------------------------------------------|
| DRDL, RCI and ASL                                      | Hyderabad      | Missile design, guidance and advanced systems      |
| ADE - Aeronautical Development Establishment           | Bengaluru      | Unmanned aerial vehicles - TAPAS, Nishant, Lakshya |
| GTRE - Gas Turbine Research Establishment              | Bengaluru      | Aero gas turbine engines - the Kaveri engine       |
| LRDE - Electronics and Radar Development Establishment | Bengaluru      | Radars                                             |
| CABS - Centre for Airborne Systems                     | Bengaluru      | Netra Airborne Early Warning and Control           |
| CVRDE - Combat Vehicles R&D Establishment              | Avadi, Chennai | Combat vehicles - the Arjun main battle tank       |
| ARDE and HEMRL                                         | Pune           | Armaments and explosives - Pinaka, ATAGS           |

| Laboratory                                         | Location                     | Area of work                                                    |
|----------------------------------------------------|------------------------------|-----------------------------------------------------------------|
| TBRL - Terminal Ballistics Research Laboratory     | Chandigarh                   | Terminal ballistics and warheads                                |
| NPOL - Naval Physical and Oceanographic Laboratory | Kochi                        | Sonar and underwater sensing                                    |
| NSTL - Naval Science and Technological Laboratory  | Visakhapatnam                | Torpedoes - Varunastra                                          |
| DFRL - Defence Food Research Laboratory            | Mysuru                       | Food technology and combat rations                              |
| INMAS, DIPAS and DIPR                              | Delhi                        | Nuclear medicine, physiology and psychological research         |
| DRDE                                               | Gwalior                      | Chemical and biological defence                                 |
| DIHAR, SASE and DIBER                              | Leh, Chandigarh and Haldwani | High-altitude agriculture, snow and avalanche study, bio-energy |

Rajasthan's own DRDO laboratory is the Defence Laboratory Jodhpur (DLJ). It works on camouflage, desert environment sciences, isotopes and radiation-detection kits - all of which follow from where it sits. Expect this in a Rajasthan-flavoured science question.

### **DRDO beyond weapons - the life-science products**

DRDO also works on keeping soldiers alive and healthy in extreme places, and those spin-offs reach civilians. This is the corner of the syllabus RPSC likes precisely because candidates skip it. Lukoskin has been asked directly.

- Lukoskin - a polyherbal drug for leucoderma (vitiligo), developed by DIBER, Haldwani, from Himalayan herbs; technology transferred to Aimil Pharmaceuticals.
- 2-deoxy-D-glucose (2-DG) - anti-COVID-19 drug developed by INMAS, Delhi, with Dr Reddy's Laboratories in 2021.
- Oxycare - an oxygen delivery system developed by DEBEL.
- Retort-pouch rations and extreme-climate food products from DFRL, Mysuru.
- Seabuckthorn-based beverages from DIHAR, Leh.
- Varunastra heavyweight torpedo from NSTL, Visakhapatnam - remember the lab, it is a common match.

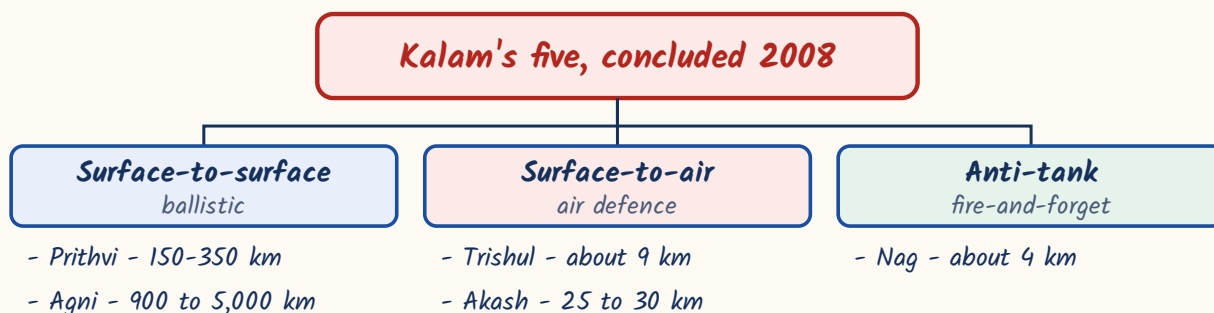
#### ASKED IN RAS

Asked in RAS Pre 2016 - Lukoskin is used in the treatment of leucoderma (vitiligo). Note the lab too: DIBER, Haldwani. RPSC has framed the same fact as both a direct question and a match.

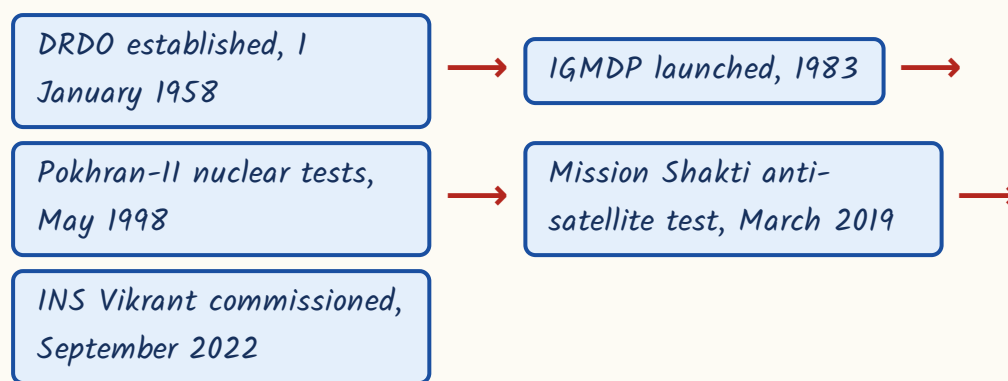
## IGMDP and the missile ladder

The Integrated Guided Missile Development Programme was launched in 1983 under A.P.J. Abdul Kalam. The idea was to build five missiles of five different kinds at once, so that India would acquire the whole set of skills rather than one. It was formally concluded in 2008, its objectives met. Learn the five names as a closed set - the exam's favourite trick is to slip a sixth name into the list.

### IGMDP 1983 - the closed set of five



### Defence technology chronology - fix these five anchors



### Missiles and their approximate ranges

| Missile    | Type                         | Approximate range  | Note                |
|------------|------------------------------|--------------------|---------------------|
| Prithvi-I  | Surface-to-surface ballistic | 150 km             | IGMDP; Army version |
| Prithvi-II | Surface-to-surface ballistic | 250 - 350 km       | Air Force version   |
| Agni-I     | Short-range ballistic        | About 700 - 900 km | IGMDP               |
| Agni-II    | Intermediate-range ballistic | 2,000+ km          |                     |
| Agni-III   | Intermediate-range ballistic | 3,000+ km          |                     |
| Agni-IV    | Intermediate-range ballistic | About 4,000 km     |                     |

| Missile            | Type                           | Approximate range | Note                                             |
|--------------------|--------------------------------|-------------------|--------------------------------------------------|
| Agni-V             | Intercontinental class         | 5,000+ km         | MIRV tested as Mission Divyashtra, 11 March 2024 |
| Agni-Prime         | Short to medium range          | 1,000 - 2,000 km  | Canisterised, newer generation                   |
| Trishul            | Short-range surface-to-air     | About 9 km        | IGMDP                                            |
| Akash              | Medium-range surface-to-air    | 25 - 30 km        | Akash-NG about 70 km                             |
| Nag                | Anti-tank guided missile       | About 4 km        | Fire-and-forget; IGMDP                           |
| QRSAM              | Quick-reaction surface-to-air  | About 30 km       | For mobile formations                            |
| VSHORADS / MANPADS | Man-portable air defence       | About 6 km        | Shoulder-fired against low-flying aircraft       |
| Astra Mk-1         | Beyond-visual-range air-to-air | 100+ km           | India's first indigenous BVR air-to-air missile  |

**ASKED IN RAS**

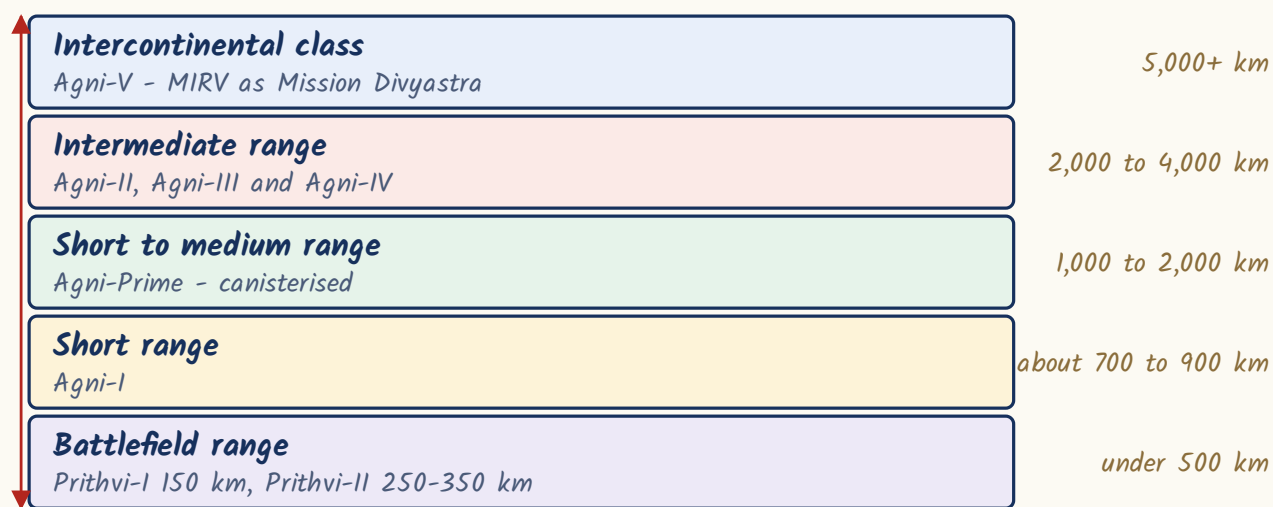
Asked in RAS Pre 2016 - which of these missiles was NOT a part of IGMDP? Answer: BrahMos. The examiner puts BrahMos among four names and waits for you to accept it as one of the five.

**ASKED IN RAS**

Two more straight repeats from this table. RAS Pre 2021 - Astra is India's first indigenous beyond-visual-range air-to-air missile. RAS Pre 2023 - MANPADS stands for a man-portable air defence system.

**YAAD RAKHO**

The IGMDP five, as one word to remember - P-A-T-N-A: Prithvi, Agni, Trishul, Nag, Akash. Range ladder for Agni: I about 900 km, II 2,000, III 3,000, IV 4,000, V 5,000 - roughly a thousand more at each step from Agni-II onwards.

**The missile range ladder - where Agni sits****TRAP**

BrahMos is NOT part of IGMDP - it is a separate India-Russia joint venture of 1998. Also do not confuse the types: Prithvi is surface-to-surface ballistic, Akash is surface-to-air, Nag is anti-tank, Astra is air-to-air. RPSC matches these four against each other almost every time.

**Cruise missiles, hypersonics and missile defence**

*A ballistic missile is thrown - it burns its engine early, arcs high above the atmosphere and falls on the target under gravity. A cruise missile flies - it stays low, keeps its engine running, and steers all the way. That single distinction answers a large family of questions, so hold it clearly.*

### **Ballistic missile versus cruise missile**

| <b>Ballistic missile</b>                                        | <b>Cruise missile</b>                                                   |
|-----------------------------------------------------------------|-------------------------------------------------------------------------|
| Powered only in the initial phase, then follows a ballistic arc | Powered and guided throughout the flight                                |
| Travels high, largely outside the atmosphere                    | Flies low, hugging the terrain to avoid radar                           |
| Very long range - Agni-V exceeds 5,000 km                       | Shorter range - BrahMos about 290 km originally, Nirbhay about 1,000 km |
| Carries larger warheads, mainly strategic                       | High accuracy, mainly tactical and precision strike                     |
| Examples: Prithvi, Agni series, K-series                        | Examples: BrahMos (supersonic), Nirbhay (subsonic)                      |

### **The systems beyond IGMDP**

| <b>System</b> | <b>Category</b>                            | <b>Key facts</b>                                                                                                                                                                                                                                                                                               |
|---------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BrahMos       | Supersonic cruise missile                  | India-Russia joint venture of 1998, named after the Brahmaputra and Moskva rivers; Mach 2.8 to 3, the world's fastest operational supersonic cruise missile; range extended beyond the original 290 km after India joined MTCR; the Philippines was the first export customer, contracted 2022, delivered 2024 |
| Nirbhay       | Subsonic cruise missile                    | Range about 1,000 km                                                                                                                                                                                                                                                                                           |
| Pinaka        | Multi-barrel rocket launcher               | Developed by ARDE, Pune; Mk-I about 38 km, enhanced about 45 km, guided Pinaka 75+ km; used in the 1999 Kargil conflict                                                                                                                                                                                        |
| Pralay        | Quasi-ballistic surface-to-surface missile | 150 - 500 km                                                                                                                                                                                                                                                                                                   |

| System           | Category                             | Key facts                                                           |
|------------------|--------------------------------------|---------------------------------------------------------------------|
| K-15<br>Sagarika | Submarine-launched ballistic missile | About 750 km - the sea leg of the nuclear triad                     |
| K-4              | Submarine-launched ballistic missile | About 3,500 km, carried by Arihant-class boats                      |
| HSTDV            | Hypersonic technology demonstrator   | Scramjet propulsion, flight tested 7 September 2020 at about Mach 6 |
| LRAShM           | Long-range hypersonic missile        | Tested from Dr A.P.J. Abdul Kalam Island in November 2024           |

- Ballistic Missile Defence Phase-I has two interceptors: the exo-atmospheric Prithvi Air Defence, which kills outside the atmosphere, and the endo-atmospheric Advanced Air Defence, which kills inside it.
- Mission Shakti, 27 March 2019, destroyed a satellite at about 283 km, making India the 4th anti-satellite power.
- A scramjet is a supersonic-combustion ramjet - air enters at supersonic speed and the fuel burns without slowing it to subsonic. No moving compressor, which is why it works only at very high speed.

## Air power - aircraft, helicopters and unmanned systems

For the exam, an aircraft question is almost always a platform-to-category match. So fix the category first - fighter, transport, helicopter, UAV, early-warning aircraft - and only then the details of who designs and who builds. Tejas is the standard example of the split: designed by the Aeronautical Development Agency, manufactured by Hindustan Aeronautics Limited.

## Aerial platforms - what each one is

| Platform                      | Category                                         | Key facts                                                                                                                     |
|-------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Tejas LCA Mk-1 / Mk-1A / Mk-2 | Light combat aircraft                            | Designed by ADA, built by HAL; IOC 2013, FOC 2019; Mk-1A powered by the GE F404; the larger Mk-2 is under development         |
| AMCA                          | Fifth-generation stealth fighter                 | India's fifth-generation programme, approved by the Cabinet Committee on Security in March 2024                               |
| Su-30MKI                      | Heavy multirole fighter                          | Licence-built by HAL at Nashik                                                                                                |
| Rafale                        | Multirole fighter                                | 36 aircraft from Dassault; first squadron No. 17 'Golden Arrows' at Ambala, 2020; 26 Rafale-M for the Navy contracted in 2025 |
| C-295                         | Medium transport aircraft                        | Assembled at Vadodara by Airbus with Tata                                                                                     |
| C-17 Globemaster and C-130J   | Heavy and special-operations transport           | Imported strategic airlift                                                                                                    |
| Dhruv ALH and Rudra           | Advanced light helicopter and its armed version  | HAL                                                                                                                           |
| LCH Prachand                  | Light combat helicopter                          | Inducted October 2022; the first unit was raised at Jodhpur                                                                   |
| HTT-40                        | Basic trainer aircraft                           | HAL                                                                                                                           |
| TAPAS-BH                      | Medium-altitude long-endurance UAV               | ADE, Bengaluru; earlier called Rustom-II                                                                                      |
| Nishant and Lakshya           | Tactical UAV and pilotless target aircraft       | ADE, Bengaluru                                                                                                                |
| SWiFT / Ghatak                | Stealth unmanned combat air vehicle demonstrator | DRDO                                                                                                                          |
| Netra AEW&C                   | Airborne early warning and control               | Built by CABS; flies on an Embraer ERJ-145                                                                                    |
| Phalcon AWACS                 | Airborne warning and control system              | Mounted on an IL-76 airframe                                                                                                  |

- The MiG-21 was retired from Indian Air Force service in September 2025 after more than six decades.
- Netra is the smaller Indian AEW&C on an Embraer; Phalcon is the larger Israeli-origin AWACS on an IL-76. Do not swap the airframes.
- The Kaveri engine, from GTRE Bengaluru, was intended for Tejas but Tejas flies on the GE F404 instead.

### TRAP

Prachand is an attack helicopter, not a trainer - the trainer is HTT-40. Dhruv is a utility helicopter and Rudra its armed version. RPSC's NOT-correctly-matched question in this section is almost always built on one of these three.

## Sea power - the Navy's platforms

Naval questions are current-affairs questions in disguise. A ship is asked about in the year it is commissioned, and the hook is always a 'first' - first indigenously designed carrier, first indigenous diving support vessel, last warship imported from abroad. Learn the ship, the year and the superlative together.

### Ships and submarines you must be able to place

| Ship             | Type                                | Commissioned     | Why it is asked                                                                                      |
|------------------|-------------------------------------|------------------|------------------------------------------------------------------------------------------------------|
| INS Vikrant      | Aircraft carrier                    | 2 September 2022 | India's first indigenously designed and built aircraft carrier; Cochin Shipyard; about 45,000 tonnes |
| INS Vikramaditya | Aircraft carrier                    | 2013             | Acquired from Russia - not indigenous                                                                |
| INS Arihant      | Nuclear ballistic missile submarine | 2016             | India's first indigenous SSBN; completed the nuclear triad with its first deterrence patrol in 2018  |

| Ship                         | Type                                               | Commissioned   | Why it is asked                                                     |
|------------------------------|----------------------------------------------------|----------------|---------------------------------------------------------------------|
| INS Arighaat                 | Nuclear ballistic missile submarine                | 2024           | Second indigenous SSBN                                              |
| INS Vagsheer                 | Kalvari-class (Scorpene) diesel-electric submarine | January 2025   | Sixth and last boat of Project 75                                   |
| INS Nilgiri                  | Project 17A stealth frigate                        | January 2025   | Lead ship of the class                                              |
| INS Surat                    | Project 15B destroyer                              | January 2025   | Visakhapatnam-class guided missile destroyer                        |
| INS Arnala                   | Anti-submarine shallow water craft                 | June 2025      | First of its class                                                  |
| INS Tamal                    | Talwar-class frigate                               | July 2025      | The last warship imported from abroad                               |
| INS Nistar                   | Diving support vessel                              | July 2025      | India's first indigenously designed and built diving support vessel |
| INS Himgiri and INS Udaygiri | Project 17A stealth frigates                       | 26 August 2025 | Commissioned together                                               |

- Project 75 - the six Kalvari-class (Scorpene) conventional submarines, built at Mazagon Dock.
- Project 15B - Visakhapatnam-class destroyers; Project 17A - Nilgiri-class stealth frigates.
- An SSBN is a nuclear-powered submarine carrying ballistic missiles; a conventional Kalvari boat is diesel-electric and cannot stay submerged as long.

**CURRENT LINK**

Current link - as of September 2026 the run of indigenous commissionings from 2025 (Nilgiri, Surat, Vagsheer, Arnala, Nistar, Himgiri, Udaygiri) is the freshest material in this chapter. If a naval question appears in RAS 2026, it will most likely be built on one of these 'firsts', or on INS Tamal as the last imported warship.

## Land systems - the Army's equipment

Group the Army's equipment by function: tanks, artillery, rockets, small arms. Then attach the developing laboratory or the manufacturer, because that is the second half of every match question here.

### Army systems, their category and their maker

| System      | Category                           | Developed or made by                        | Note                                   |
|-------------|------------------------------------|---------------------------------------------|----------------------------------------|
| Arjun       | Main battle tank                   | CVRDE, Avadi                                | India's indigenous MBT                 |
| Zorawar     | Light tank                         | DRDO with Larsen and Toubro                 | For high-altitude terrain              |
| ATAGS       | 155 mm / 52 calibre towed howitzer | ARDE, Pune                                  | Range about 48 km                      |
| Dhanush     | 155 mm / 45 calibre towed howitzer | Ordnance factories, now AWEIL               | Called the 'Desi Bofors'               |
| K-9 Vajra-T | Tracked self-propelled gun         | L&T with Hanwha                             | Mobile artillery for desert and plains |
| M777        | Ultra-light howitzer               | Imported from BAE Systems                   | Air-portable, for mountain formations  |
| Pinaka      | Multi-barrel rocket launcher       | ARDE, Pune                                  | Used in the Kargil conflict, 1999      |
| AK-203      | Assault rifle                      | Indo-Russian joint venture at Korwa, Amethi | Replacing the INSAS rifle              |

- INSAS was the earlier indigenous rifle family; AK-203 is its replacement, made in India at Korwa in Amethi district, Uttar Pradesh.
- Varunastra (torpedo, NSTL) and Arjun (tank, CVRDE) are the two lab-to-product pairs RPSC has actually used.
- Note the calibre difference: ATAGS is 155 mm / 52 cal, Dhanush is 155 mm / 45 cal. The calibre number after the slash is barrel length in calibres, and a longer barrel means longer range.

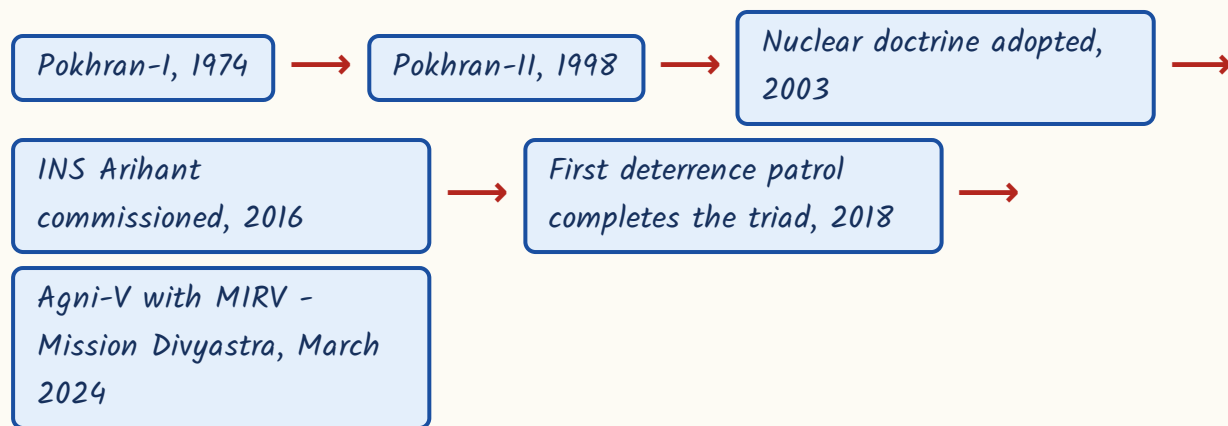
## Nuclear India - tests, doctrine, triad and the export-control regimes

India's nuclear story has a Rajasthan address. Both test series were conducted at Pokhran, in Jaisalmer district. Pokhran-I in 1974 was called a 'peaceful nuclear explosion'; Pokhran-II in 1998 was an open declaration of weapon capability, and it brought sanctions that shaped the next two decades of Indian technology policy.

### Pokhran-I versus Pokhran-II

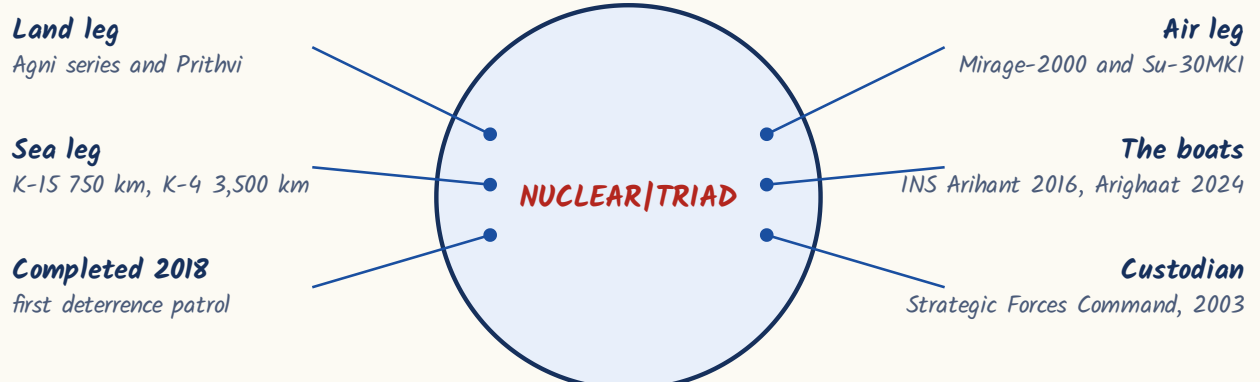
| Pokhran-I - Smiling Buddha                                 | Pokhran-II - Operation Shakti                     |
|------------------------------------------------------------|---------------------------------------------------|
| 18 May 1974                                                | 11 and 13 May 1998                                |
| A single device, described as a peaceful nuclear explosion | Five devices, including a thermonuclear design    |
| Prime Minister Indira Gandhi                               | Prime Minister Atal Bihari Vajpayee               |
| Led to the creation of the Nuclear Suppliers Group in 1975 | 11 May is now observed as National Technology Day |
| No formal doctrine followed                                | Doctrine formally adopted in 2003                 |

### How the nuclear triad was completed



- The triad means the ability to deliver nuclear weapons from land, air and sea. India's legs are the Agni missiles, aircraft such as the Mirage-2000 and Su-30MKI, and the K-series missiles on Arihant-class submarines.
- Doctrine of 2003 - No First Use; credible minimum deterrence; massive retaliation if attacked; no use against non-nuclear weapon states.
- The Strategic Forces Command, created in 2003, manages the nuclear arsenal.
- Mission Divyastra, 11 March 2024, was the first Indian flight test of Agni-5 with Multiple Independently Targetable Re-entry Vehicles - one missile, several warheads aimed at different targets.

### The nuclear triad - completed in 2018



### Export-control regimes - what India is in and what it is not

| Regime                                   | India's status | Year |
|------------------------------------------|----------------|------|
| MTCR - Missile Technology Control Regime | Member         | 2016 |
| Wassenaar Arrangement                    | Member         | 2017 |
| Australia Group                          | Member         | 2018 |
| NSG - Nuclear Suppliers Group            | NOT a member   | -    |
| NPT - Non-Proliferation Treaty           | Not signed     | -    |
| CTBT - Comprehensive Test Ban Treaty     | Not signed     | -    |

**TRAP**

India is in three of the four regimes but NOT in the Nuclear Suppliers Group, and has signed neither the NPT nor the CTBT. RPSC's statement-based question here usually plants NSG membership as the false statement. Second confusion: Pokhran-I was 1974 under Indira Gandhi, Pokhran-II 1998 under Vajpayee - National Technology Day, 11 May, commemorates the 1998 tests, not 1974.

## Defence production, exports, joint exercises and the tri-services structure

The last decade's story is indigenisation - build at home, then sell abroad. Three structural changes carry it: the ordnance factories were corporatised, two defence corridors were created, and start-ups were brought in through iDEX. The numbers move every year, so carry the latest figure with a date attached.

- The Ordnance Factory Board was dissolved on 1 October 2021 and its units reorganised into 7 new DPSUs - Munitions India, Armoured Vehicles Nigam (AVANI), Advanced Weapons and Equipment India, Troop Comforts, Yantra India, India Optel and Gliders India.
- Two defence industrial corridors were announced in Budget 2018-19 - Uttar Pradesh (Aligarh, Agra, Jhansi, Kanpur, Chitrakoot, Lucknow) and Tamil Nadu (Chennai, Coimbatore, Hosur, Salem, Tiruchirappalli).
- iDEX (2018) engages start-ups and MSMEs through the Defence Innovation Organisation.
- The SRIJAN portal and the Positive Indigenisation Lists support Atmanirbharta by barring the import of listed items after fixed dates.
- Defence exports reached about Rs 23,622 crore in 2024-25, against a target of Rs 50,000 crore by 2029.

#### CURRENT LINK

Current link - India's defence exports rose to an all-time high of about Rs 38,424 crore in 2025-26, up roughly 63 per cent on the previous year's Rs 23,622 crore, with DPSUs contributing about 54.8 per cent and the private sector about 45.2 per cent (Ministry of Defence figures released April 2026). If RPSC asks a defence-export number in 2026, this is the one to carry - always with the financial year attached.

Now the joint exercises. This is asked almost every year, and it is pure memorisation, so treat it as a vocabulary list. Two tables below; learn the exercise name to the country, and note which service it belongs to, because RPSC sometimes asks that too.

#### Joint military exercises - army and bilateral (part I)

| Exercise    | Partner country          | Service |
|-------------|--------------------------|---------|
| Yudh Abhyas | United States of America | Army    |

| <b>Exercise</b>  | <b>Partner country</b>   | <b>Service</b>           |
|------------------|--------------------------|--------------------------|
| Vajra Prahar     | United States of America | Special Forces           |
| Indra            | Russia                   | Bilateral, services-wise |
| Surya Kiran      | Nepal                    | Army                     |
| Sampriti         | Bangladesh               | Army                     |
| Mitra Shakti     | Sri Lanka                | Army                     |
| Ekuverin         | Maldives                 | Army                     |
| Dustlik          | Uzbekistan               | Army                     |
| Nomadic Elephant | Mongolia                 | Army                     |
| Al Nagah         | Oman                     | Army                     |
| Cyclone          | Egypt                    | Special Forces           |
| Maitree          | Thailand                 | Army                     |
| Dharma Guardian  | Japan                    | Army                     |
| Harimau Shakti   | Malaysia                 | Army                     |

### Joint military exercises - naval, air and multilateral (part 2)

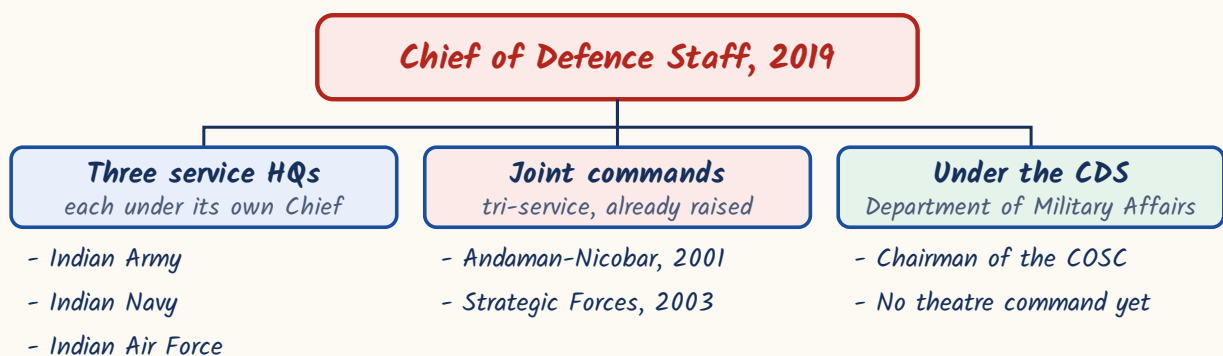
| <b>Exercise</b> | <b>Partner country</b>   | <b>Service</b> |
|-----------------|--------------------------|----------------|
| Malabar         | USA, Japan and Australia | Navy           |
| Varuna          | France                   | Navy           |
| Garuda          | France                   | Air Force      |
| Shakti          | France                   | Army           |
| Konkan          | United Kingdom           | Navy           |
| Ajeya Warrior   | United Kingdom           | Army           |
| SIMBEX          | Singapore                | Navy           |
| SLINEX          | Sri Lanka                | Navy           |
| JIMEX           | Japan                    | Navy           |
| AUSINDEX        | Australia                | Navy           |
| AUSTRA HIND     | Australia                | Army           |
| Cope India      | United States of America | Air Force      |

| Exercise          | Partner country | Service |
|-------------------|-----------------|---------|
| Al Mohed Al Hindi | Saudi Arabia    | Navy    |
| Samudra Shakti    | Indonesia       | Navy    |

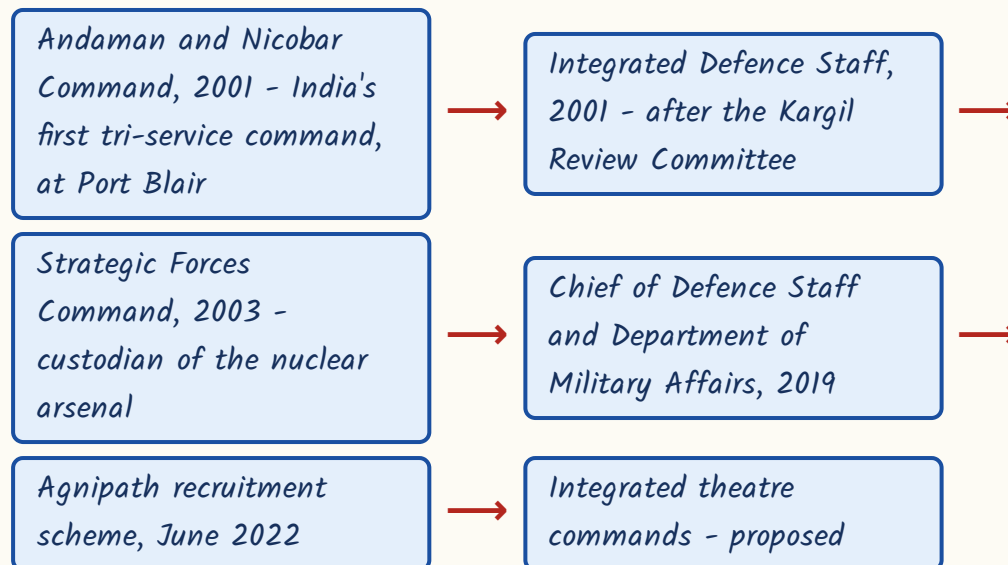
**TRAP**

Four swaps carry almost every question in this section. France has three exercises - Varuna (navy), Garuda (air force) and Shakti (army). Al Nagah is with Oman, not Egypt; the India-Egypt exercise is Cyclone. Malabar is a four-nation naval exercise of India, the USA, Japan and Australia, not a bilateral. And in the structural set: Agnipath (June 2022) is a recruitment scheme, NOT the corporatisation of the ordnance factories - that was 2021; the Integrated Defence Staff dates from 2001, while 2019 is the year of the CDS and the Department of Military Affairs.

Finally, the command structure. India's forces are still organised service by service, but a set of tri-service bodies has been layered on top to make them fight as one. Learn the years - RPSC has asked exactly this as a NOT-correct question.

**The tri-service structure**

### Building the tri-services structure



- The post of Chief of Defence Staff was created in 2019. General Bipin Rawat became the first CDS on 1 January 2020; General Anil Chauhan has held the post since September 2022.
- The CDS heads the Department of Military Affairs and is the Permanent Chairman of the Chiefs of Staff Committee.
- Agnipath, launched June 2022 - a four-year Agniveer tenure with up to 25 per cent retained in regular service.
- As of September 2026 no integrated theatre command had been formally raised; the CDS submitted a final plan to the Defence Ministry in 2026 and it awaits government clearance.
- Rajasthan and defence: the Pokhran Field Firing Range (nuclear tests, Exercise Bharat Shakti 2024, the IAF's Vayu Shakti firepower demonstration), the Mahajan Field Firing Range near Bikaner, the Chandan range near Jaisalmer, Defence Laboratory Jodhpur, and the Jodhpur air base which hosted a phase of the multinational Exercise Tarang Shakti in 2024.

### Central armed police forces - the raising years RPSC asks

| Force                      | Raised | Note                                   |
|----------------------------|--------|----------------------------------------|
| Assam Rifles               | 1835   | The oldest paramilitary force in India |
| Indo-Tibetan Border Police | 1962   | Raised after the India-China war       |

| Force                             | Raised | Note                                        |
|-----------------------------------|--------|---------------------------------------------|
| Central Industrial Security Force | 1969   | Guards industrial and airport installations |
| Border Security Force             | 1965   | India's largest border guarding force       |
| National Security Guard           | 1984   | Counter-terrorism force - not 1974          |

## REVISION RECAP

1. DRDO - founded 1 January 1958, headquarters DRDO Bhawan New Delhi, about 52 laboratories, motto Balasya Mulam Vigyanam from the Arthashastra.
2. Lab-to-city: Hyderabad missiles, Bengaluru aeronautics and radar, Pune armaments, Kochi sonar, Visakhapatnam torpedoes, Mysuru food, Avadi combat vehicles.
3. Rajasthan's DRDO lab is Defence Laboratory Jodhpur - camouflage, desert sciences, radiation detection.
4. Lukoskin for leucoderma from DIBER Haldwani; 2-DG anti-COVID drug from INMAS Delhi; Oxycare from DEBEL; rations from DFRL Mysuru.
5. IGMDP - launched 1983 under A.P.J. Abdul Kalam, concluded 2008; the five are Prithvi, Agni, Trishul, Akash and Nag. BrahMos is not one of them.
6. Ranges: Prithvi-I 150 km, Prithvi-II 250-350 km, Agni-I about 900 km, Agni-II 2,000+, Agni-III 3,000+, Agni-IV about 4,000, Agni-V 5,000+, Agni-Prime 1,000-2,000 km.
7. Types: Prithvi surface-to-surface, Akash surface-to-air, Nag anti-tank, Astra beyond-visual-range air-to-air, QRSAM quick-reaction SAM.
8. BrahMos - India-Russia joint venture 1998, named after the Brahmaputra and Moskva, Mach 2.8-3; the Philippines was the first export customer.
9. Mission Divyasthra, 11 March 2024 - first Agni-5 test with MIRV. HSTDV, September 2020 - scramjet at about Mach 6. Mission Shakti, March 2019 - anti-satellite kill at about 283 km.
10. Ballistic missile arcs unpowered after boost; cruise missile flies low under power all the way. BMD Phase-I = exo-atmospheric PAD plus endo-atmospheric AAD.
11. Tejas - designed by ADA, built by HAL, IOC 2013 and FOC 2019, Mk-1A on the GE F404. AMCA is the fifth-generation programme cleared by the CCS in March 2024.
12. Netra AEW&C on an Embraer ERJ-145 from CABS; Phalcon AWACS on an IL-76. TAPAS-BH from ADE Bengaluru. Prachand is the light combat helicopter, first unit at Jodhpur.
13. INS Vikrant, September 2022 - first indigenously designed and built carrier; Vikramaditya was bought from Russia in 2013.
14. Arihant 2016 and Arighaat 2024 are the indigenous SSBNs; Vagsheer, January 2025, closed Project 75; INS Nistar is the first indigenous diving support vessel; INS Tamal is

the last imported warship.

15. Arjun from CVRDE Avadi, Pinaka and ATAGS from ARDE Pune, Varunastra from NSTL Visakhapatnam, AK-203 from Korwa in Amethi.
16. Pokhran-I 'Smiling Buddha' 18 May 1974, Pokhran-II 'Operation Shakti' 11 and 13 May 1998 - both in Jaisalmer district; 11 May is National Technology Day.
17. Doctrine 2003 - No First Use, credible minimum deterrence, massive retaliation, no use against non-nuclear weapon states; triad completed 2018 by INS Arihant's first deterrence patrol.
18. MTCR 2016, Wassenaar 2017, Australia Group 2018 - but NOT the NSG, and neither NPT nor CTBT signed.
19. OFB dissolved 1 October 2021 into 7 DPSUs; defence corridors in Uttar Pradesh and Tamil Nadu; exports about Rs 38,424 crore in 2025-26 against a Rs 50,000 crore target for 2029.
20. CDS created 2019, Bipin Rawat first from 1 January 2020, Anil Chauhan since September 2022; Andaman and Nicobar Command 2001 was the first tri-service command; Agnipath from June 2022 with four-year tenure and up to 25 per cent retention.